

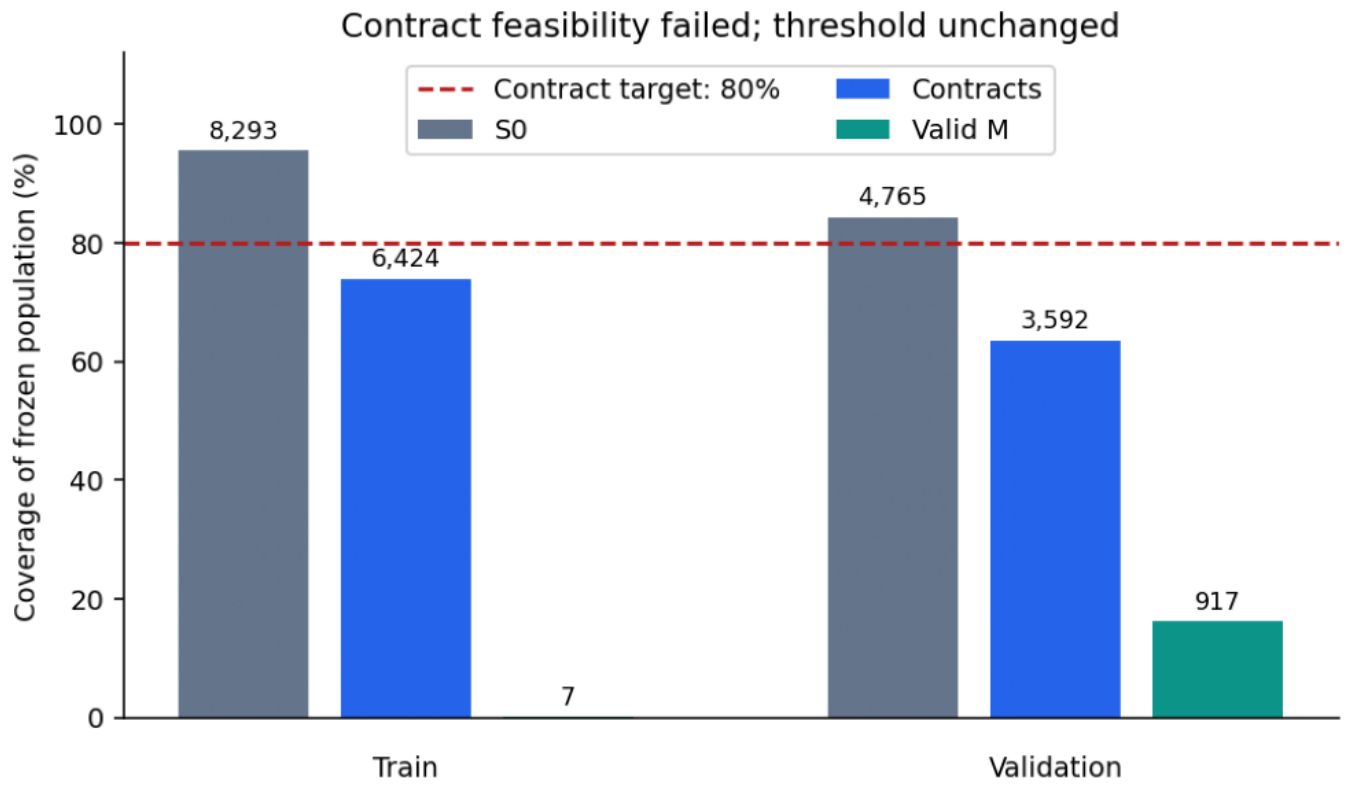
Earnings moves beyond the implied range

Preregistered continuation hypothesis; transparent feasibility study and exploratory implementation demonstration. The historical-contract feasibility gate FAILED in both development splits. We preserve the 80% threshold and do not replace failed events or claim confirmatory strategy success.

Economic rationale: a completed earnings reaction exceeding the pre-event ATM straddle-premium move proxy may reflect incomplete information incorporation. Overreaction is an alternative; remaining option time value and risk premia confound the proxy. Historical information, fixed windows and simple interpretable rules avoid future-price signal inputs.

S0 is the median valid pre-event underlying NBBO midpoint. Select the earliest standard unadjusted 100-share expiry from D+7 through D+21 inclusive, nearest matching call/put strike, lower strike on a tie. M is the median straddle-midpoint/underlying ratio at at least three shared valid completed minutes in [15:50,15:55) ET. Positive two-sided prices/sizes, unlocked/uncrossed books, <=20% relative option spreads and valid feed timestamps are required.

Split	Population	Valid S0	Pairs	Contract %	Valid M
train	8688	8293	6424	73.94%	7
validation	5652	4765	3592	63.55%	917



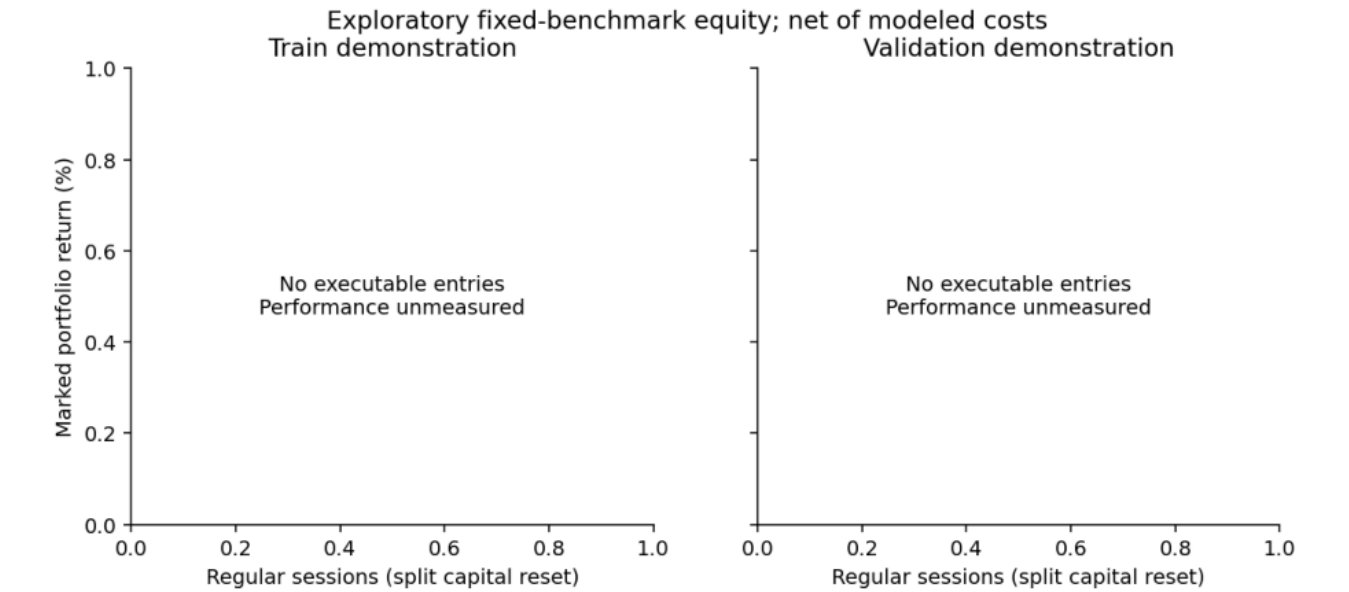
Frozen trading and cost implementation

At D, complete stock/SPY reaction measurement in [10:00,10:05) ET. r is the split-comparable stock price reaction; a is its total-return reaction minus clipped beta times SPY total-return reaction. Beta is intercept OLS over the preceding 120 sessions with ≥ 100 paired returns, clipped to [0,2]. $X = \text{abs}(r)/M$; require finite nonzero sign-consistent r and a . Trade only $X > 1$ in $\text{sign}(a)$.

Enter in [10:10,10:20) ET; exit in the same window ten regular sessions later, $D = \text{session zero}$. Size 1% of initial \$1 million reference equity per issuer, 50% aggregate stock allocation cap, proportional simultaneous sizing, fixed shares except actions. Known S1 sets order intentions. All same-security orders share 1% observed execution-window volume. Missing/partial scheduled exits remain unresolved exposures.

Modeled fills pay half median NBBO spread plus 2 bps adverse slippage and \$0.001/share commission. Report doubled modeled friction and 0.005/*share*/1-min commissions separately. Historical sell-side SEC/FINRA fees and action cashflows are included; exchange/CAT route pass-through remains unresolved. Missing historical borrow excludes shorts and short hedge legs from executable returns, not research observations. Reference fills are benchmarks, not guaranteed execution.

Split	Implementation	Entries	Ann. return	Sharpe	Max DD
train	directional	0	unmeasured	unmeasured	unmeasured
train	doubled_cost	0	unmeasured	unmeasured	unmeasured
train	conserv. comm.	0	unmeasured	unmeasured	unmeasured
train	physical_hedge	0	unmeasured	unmeasured	unmeasured
validation	directional	0	unmeasured	unmeasured	unmeasured
validation	doubled_cost	0	unmeasured	unmeasured	unmeasured
validation	conserv. comm.	0	unmeasured	unmeasured	unmeasured
validation	physical_hedge	0	unmeasured	unmeasured	unmeasured

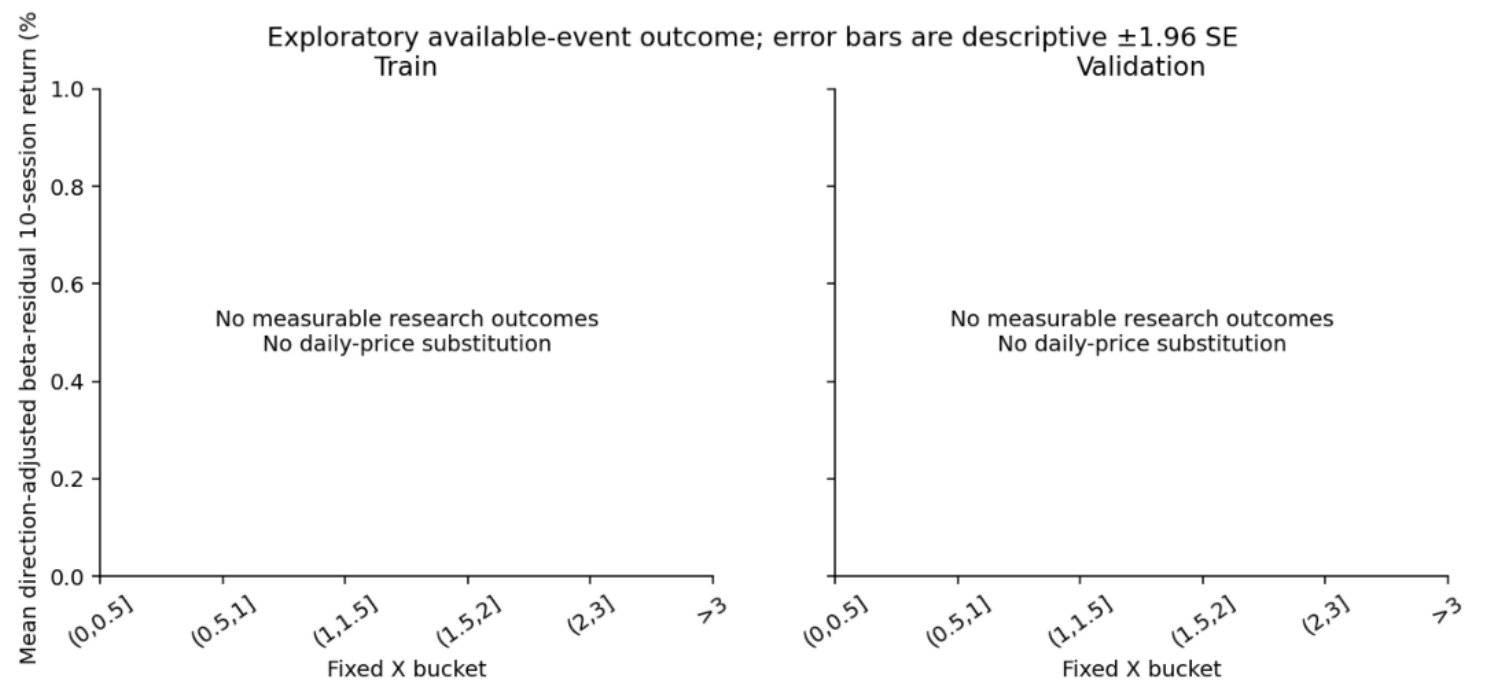


Available-event outcomes and limitations

The full-universe S0/contracts/M population remains intact. The fixed outcome-blind 20-event NBBO benchmark is only an exploratory downstream demonstration because the shared request ceiling cannot fund full-universe reaction/outcome windows. No failed event is backfilled. Calendar boundary crossings are purged before outcome analysis.

Research Y is direction-adjusted stock-minus-beta-SPY total return between prescribed NBBO entry/exit references. Both X groups remain eligible regardless of borrow, allocation or execution. Primary OLS uses $I(X>1)$, $\text{abs}(r)$, and RV20 with two-way issuer/date clustering. Missing groups/rank deficiency are inconclusive; fewer than 30 clusters in either dimension is unreliable. No outcome-based thresholds, trimming or tuning.

Split	Fixed events	Signal measurable	Outcomes	X>1	X<=1
train	10	0	0	0	0
validation	10	0	0	0	0



Holdout: sealed_pending_implementation_freeze. A durable implementation/specification hash freeze precedes 2026 access. The runner permits one attempt only; missing validated funded inputs yield an incomplete/inconclusive result, never fabricated returns or new dates.

Limitations: FMP original-release/vendor vintage and uninterrupted common-stock status between reference brackets are unproven. Minute sampling does not prove continuous liquidity. Complex corporate actions, exchange/CAT costs, historical borrow and execution attainability constrain interpretation. Full CSV ledgers, metrics, frozen source hashes and source links accompany this PDF in results/final/ and README.md.